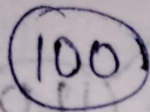
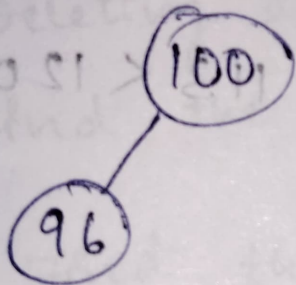


29) Create a BST using the following values..
100, 96, 110, 98, 95, 120, 109, 113, 130, 83

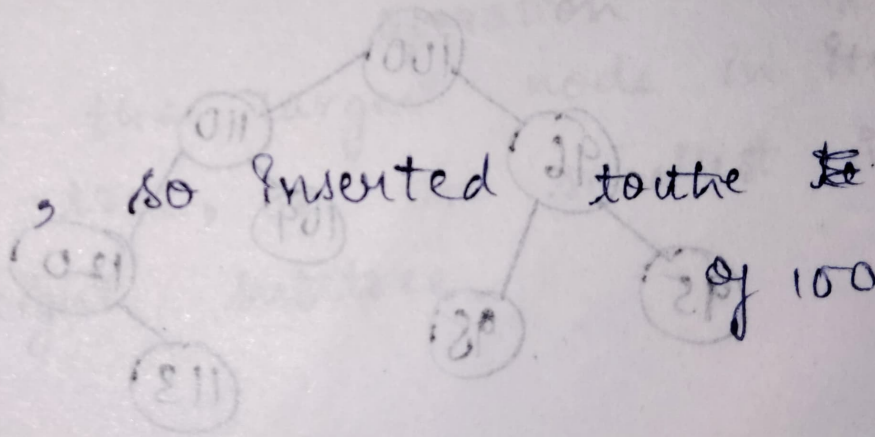
⇒ 100 is the very first value so it is first parent node or root node..

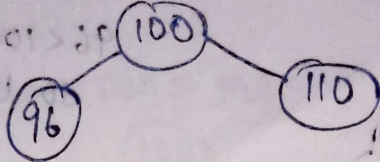


⇒ $96 < 100$, so left of 100. It is inserted.

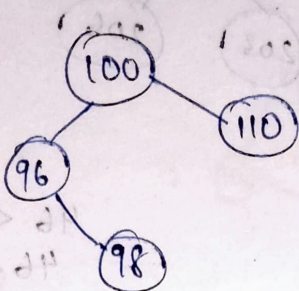


⇒ $110 > 100$, so inserted to the right of 100

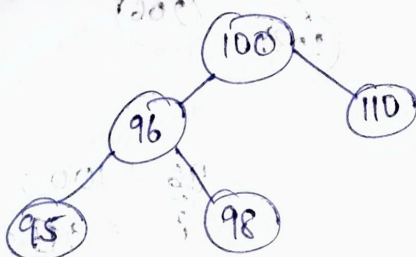




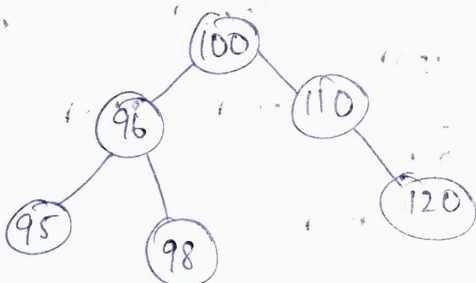
$\Rightarrow 98 < 100$; $98 > 96$, so inserted towards right of 96.



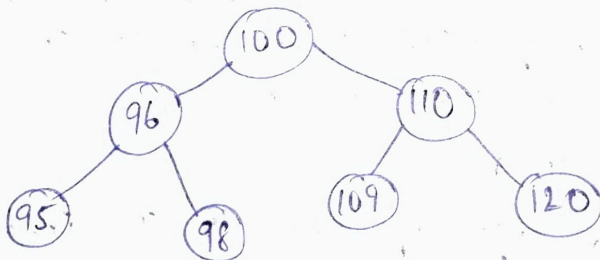
$\Rightarrow 95 < 100$; $95 < 96$; so inserted towards left of 96.



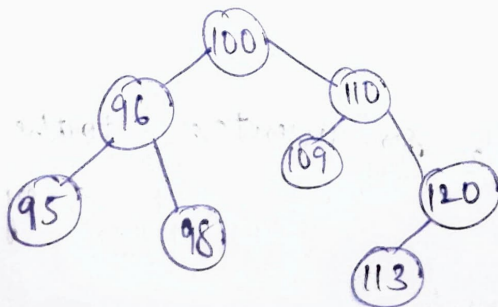
$\Rightarrow 120 > 100$; $120 > 110$, so inserted right of 110.



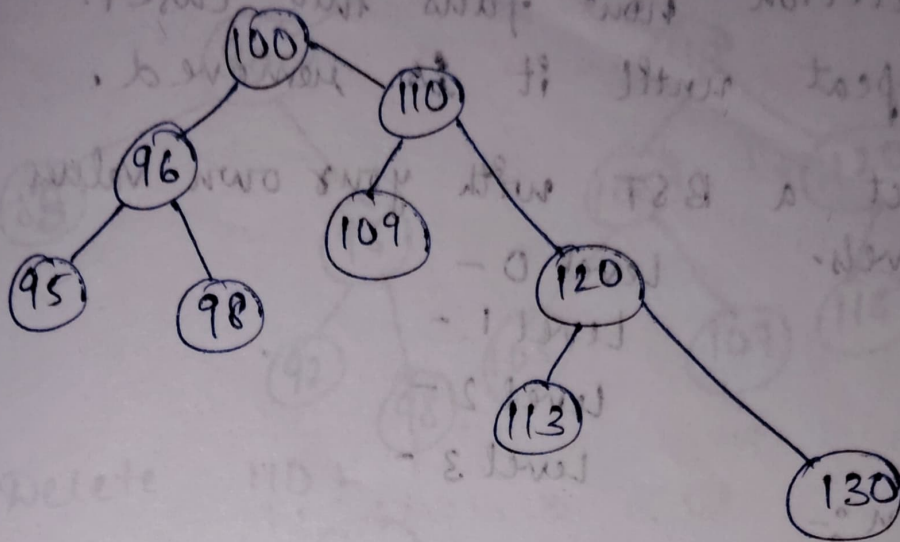
$\Rightarrow 109 > 100$; $109 < 110$, so inserted left of 110



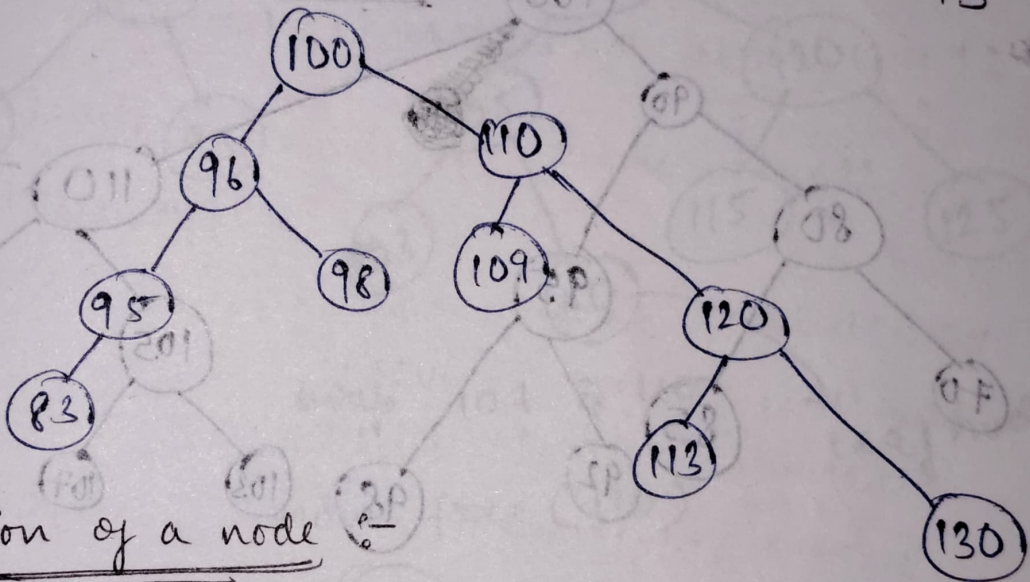
$\Rightarrow 113 > 100$; $113 > 110$; $113 < 120$, so left of 120.



$\Rightarrow 130 > 100$; $130 > 110$; $130 > 120$, so right of 120



$\Rightarrow 83 < 100$; $83 < 96$; $83 < 95$, so left of 95



Deletion of a node

To delete a node there are 3 cases:

Case 1: Deleting a leaf node:

- $\rightarrow$ find the node using search operation
- $\rightarrow$ free it and stop - nothing else is affected

Case 2: Deleting a node with one child:

- $\rightarrow$ find the node using the search operation
- replace the node directly to its only child.

Q. Construct a BST with your own values having 4 levels.

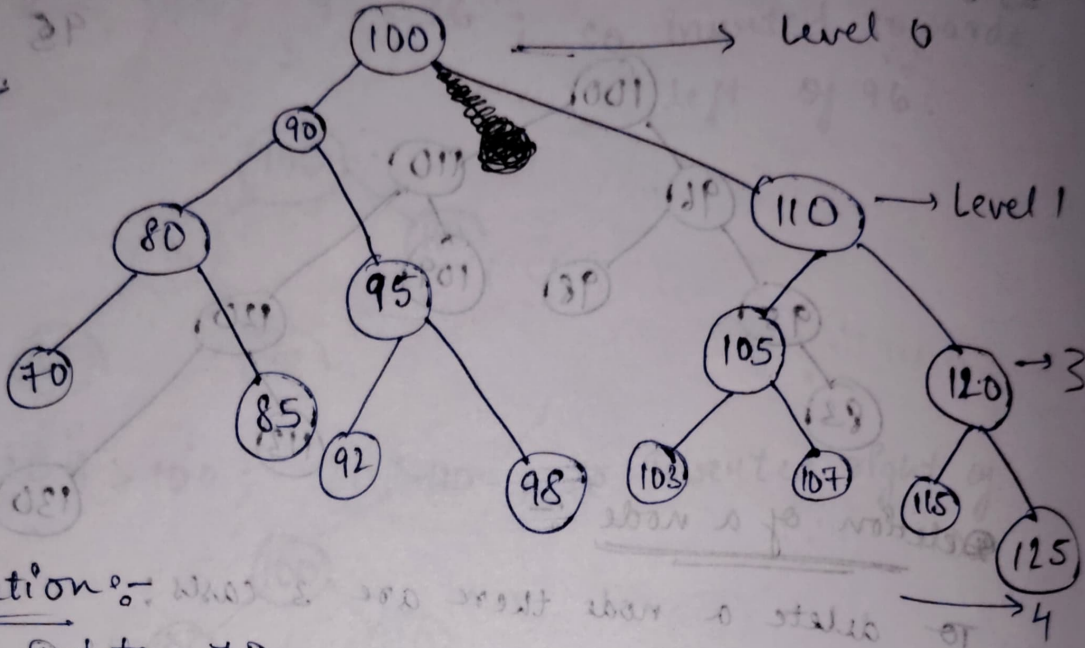
Level 0 -

Level 1 -

Level 2 -

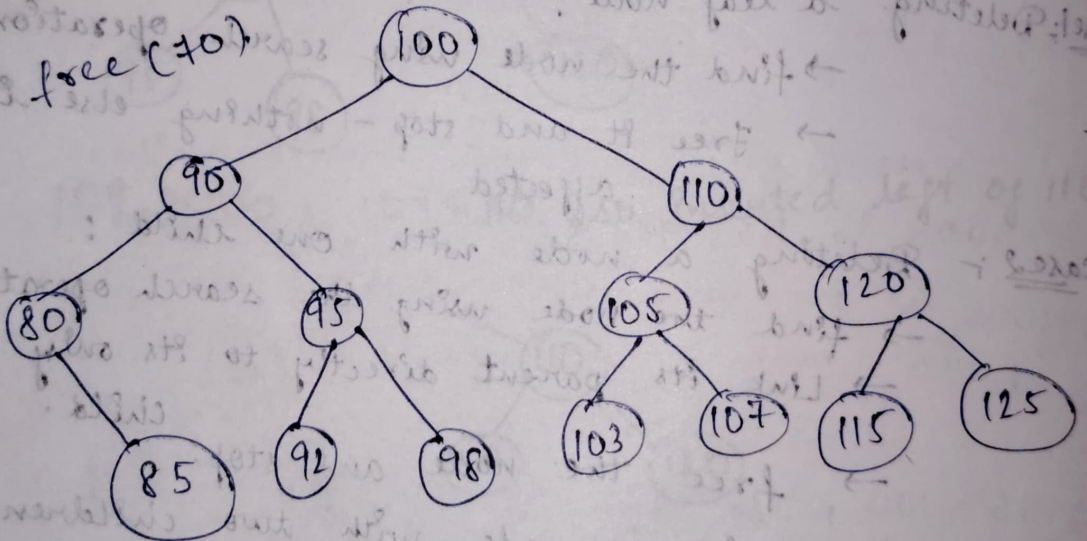
Level 3 -

Insertion:-

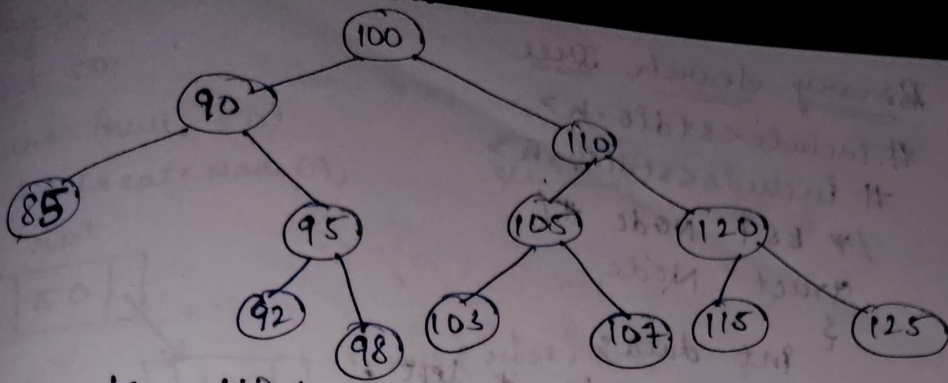


Deletion:-

(i) Delete 70.

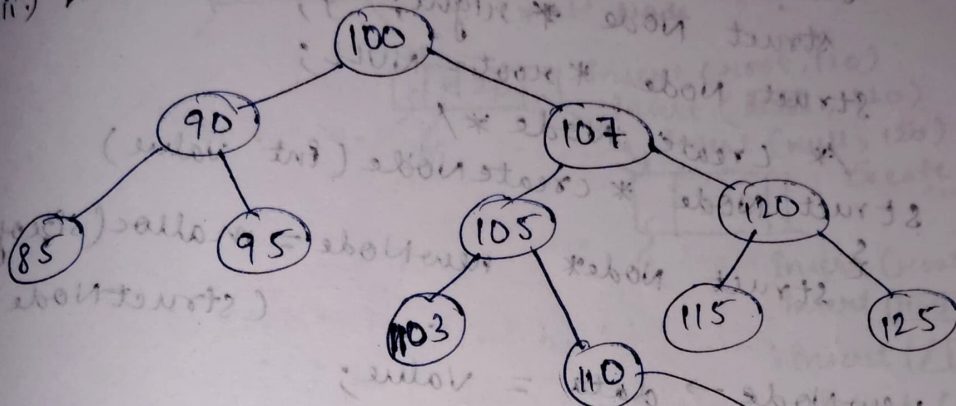


(ii) Delete 85 :-



88% Delete

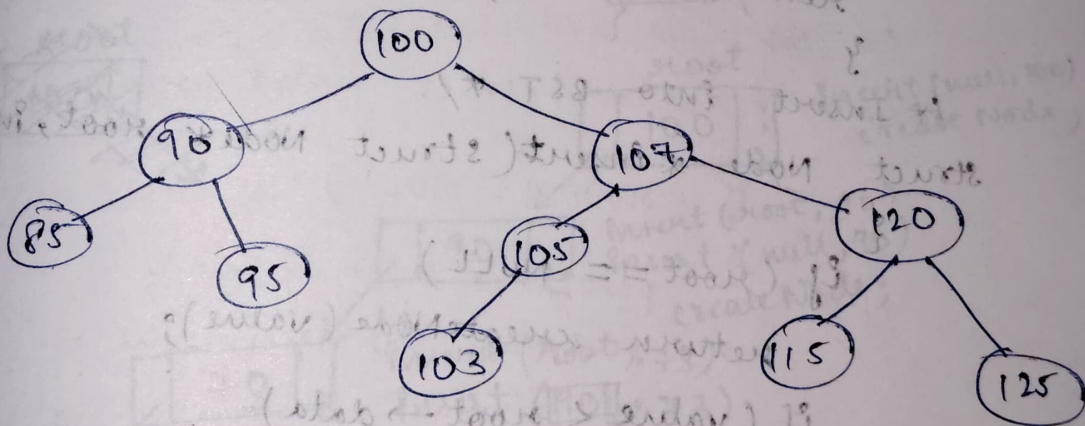
110 :-



swap 107 & 110.

leaf node.

now free (110)



In order : [LPR] $\Rightarrow$ 70, 80, 85, 90, 92, 95, 98, 100, 103, 105, 107, 110, 115, 120, 125

Preorder : [PLR] $\Rightarrow$ 100, 90, 80, 70, 85, 95, 92, 98, 110, 105, 103, 107, 120, 115, 125.

Post order : [LRP] $\Rightarrow$ 70, 85, 80, 92, 98, 95, 90, 103, 107, 105, 115, 125, 120, 110, 100